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Coordinating Curriculum Implementation Using Wiki-supported Graph Visualization

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Abstract

Traditionally, curriculum specifications tend to be formal, static documents, not encouraging adaptation and evolution. In order to facilitate various activities in implementing a curriculum, such as coordinating content, meta-cognitive competences, student-centered learning outcomes, this work proposes to employ a structured, Wiki-supported space. Our experience with this approach showed an improved transparency and coordination of courses based on a structured overview, a multi-faceted specification and increased sharing among teaching staff.

1. Active Curriculum of Computer Science

The Active Curriculum of Computer Science (ActiveCC) is a strategic project of the Faculty of Computer Science at the University of Vienna. The main goal of the project is to facilitate transparency, coordination and sharing of knowledge among teaching staff in order to implement a well-coordinated and coherent new computer science bachelor curriculum. The visualization and multi-faceted specification of the curriculum as described in this work is one special "ingredient" of the ActiveCC project.

2. Visualization of the Curriculum

According to the structural specifications stated in the formal curriculum document we organized the curriculum's Wiki offered on the CEWebS (Cooperative Environment Web Services) platform [1] in three layers: the curriculum, module and course layer. The curriculum layer includes a graph visualization of the structure of the curriculum (as illustrated in Figure 1) as well as general information like the curriculum goals, qualification profile of graduates, academic degree of graduates, etc.

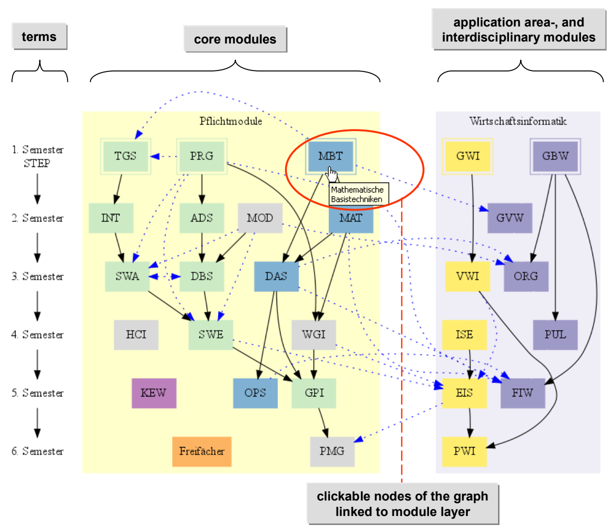

Figure 1: Visualization of curriculum structure

The graph shown in Figure 1 illustrates the modules of the curriculum and their dependencies. The organization of the graph relates to the suggested chronological order of the study. The position of a module within the graph complies with the semester in which its courses take place. For example, the courses of the module "MBT" (Basic Techniques in Mathematics) – which is positioned in the first row of the graph - are held in winter term and are suggested to be attended in the first semester by the students. The color scheme communicates the affiliation of the module to a particular module group. For example, green modules are members of the information technology module group, blue modules are considered as structural science modules, interdisciplinary modules are labeled in yellow, and violet modules represent application area modules. The arrows among the modules are dependencies that are subdivided into those stated in the formal curriculum document (black arrows) and those related to the modules/courses content (blue dotted arrows). The content related dependencies were identified by teaching staff during

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interviews. A more detailed description of the content-related dependencies is given on the module layer (illustrated in Figure 2).

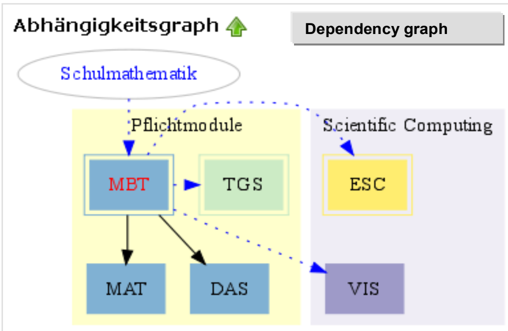

Figure 2: A module's dependency graph

The module layer contains module descriptions that are based on a description template. Items of the template are for example the module title, module coordinator, subject-specific and generic learning goals, content, methods, assessment mode and links to the module's courses. Further items are a dependency graph which illustrates direct dependencies of the module with other modules (Figure 2) as well as a description of dependencies in terms of prerequisites and dependent modules. The content-related prerequisites are described by a list of content items of other modules that are based on the educator's perceived necessity to be taught before, or in some cases simultaneously, to the course(s) of the module.

At the course layer, courses of the modules are described in Wiki pages by means of a description template that includes the items course's title, type, credits, content, methods, assessment mode and literature. Methods of the course are textually explained and visualized by coUML activity diagram graphs [2]; see an example in Figure 3.

3. Participatory Curriculum Design

In interviews that referred to curriculum, module and course issues we asked our teaching staff (N=55, n=32) to describe their perception of ActiveCC's usefulness. Results of the content analysis show that educators see benefits in:

- The visualization of the curriculum as a graph as it gives an overview of the curriculums structure and shows the position of the module within the curriculum. The illustration of dependencies among modules helps educators to get an overview of the links between their module and other modules of the curriculum. Thus, the role of the module in the curriculum becomes more transparent.
- The transparency of other modules' and courses' content as this information helps them to plan their own courses, to coordinate content with other modules in order to avoid unintended content redundancies or the absence of important topics. On the basis of the module and course descriptions content-related dependencies could be easily identified by the educators.

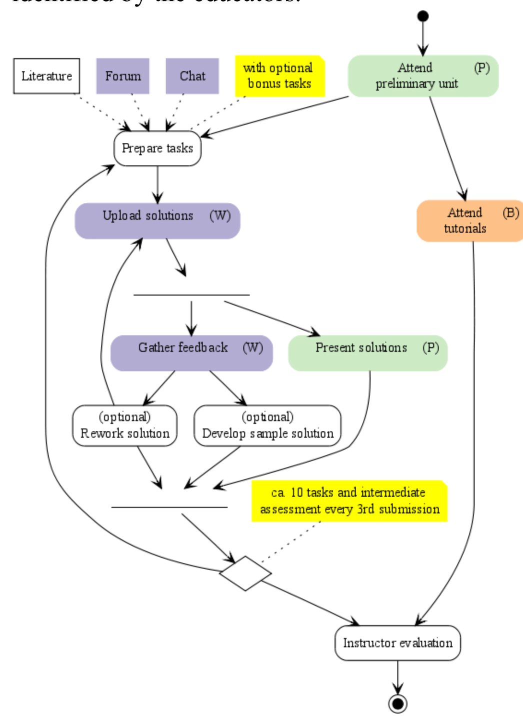

Figure 3: A course's activity diagram graph

By showing our teaching staff the curriculum visualizations and module and course descriptions many content-related dependencies as well as a course that did not fit with its position in the curriculum could be identified. Additionally, coordination and exchange among teaching staff was initiated and facilitated.

References

[1] J. Mangler, "CEWebS - Cooperative Environment Web Services," Faculty of Computer Science, University of Vienna, Master thesis, 2005.

[2] M. Derntl and R. Motschnig-Pitrik, "coUML – A Visual Language for Modeling Cooperative Environments," in Handbook of Visual Languages for Instructional Design: Theories and Practices, L. Botturi and T. Stubbs, Eds. Hershey, PA: Information Science Reference, 2007, pp. 155-184.